Generalised Linear Models from Scratch for Insurance Pricing

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1 Why This Question?

Insurance pricing is fundamentally a prediction problem: given what we know about a policyholder, how much should we expect their claims to cost? The standard actuarial approach uses *Generalised Linear Models* (GLMs), which are required by regulators across Europe for technical pricing. As an aspiring actuary, the connection between *Statistical Modelling I* and real industry practice felt natural, and with the opportunity of this coursework, I wanted to build a GLM from scratch and apply it to a real pricing problem.

In class, we developed linear models from first principles: defining the model (§9.4), deriving the least squares estimator (§9.6–9.7), proving its properties via the Gauss–Markov theorem (§9.7), and building tests and confidence intervals under Normal Theory Assumptions (§10). However, insurance data violates the assumptions underpinning this framework. Claim counts are non-negative integers, claim amounts are positive and right-skewed, and the variance depends on the mean. The brief introduction to GLMs in §11.9, while not covered in lectures, hinted at a resolution but left the theory largely undeveloped.
This project addresses that gap. We extend the theory to the GLM framework, show how the estimation procedure connects back to techniques already covered in the course, and apply the full methodology to an insurance dataset. It is less clear whether the diagnostics developed for linear models in §11 transfer cleanly to the GLM setting, and this is something we return to in Section 4.3.

2 From Linear Models to GLMs

2.1 Why the Linear Model Fails for Insurance Data

Recall the linear model from §9.4: $\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$, where $\mathrm{E}(\boldsymbol{\epsilon}) = \mathbf{0}$ and $\mathrm{Cov}(\boldsymbol{\epsilon}) = \sigma^2 I_n$ (Second Order Assumptions). Under Normal Theory Assumptions (§10), we additionally require $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma^2 I_n)$.
Several assumptions break for insurance data, and they are related.
(i) Non-normal responses and the linearity problem. Claim counts are non-negative integers, naturally modelled by a Poisson distribution. Claim sizes are positive and right-skewed. Neither is Gaussian, so the t-tests (§10.3), F-tests (§10.4), and exact confidence intervals from §10 are invalid. This is tied to the fact that a linear predictor $X\boldsymbol{\beta}$ can produce negative fitted values for counts, which is nonsensical – the response space and the model space do not match.
(ii) Non-constant variance. For Poisson data, $\mathrm{Var}(Y_i) = \mathrm{E}(Y_i)$: the variance depends on the mean. This violates the SOA $\mathrm{Cov}(\mathbf{Y}) = \sigma^2 I_n$. This is exactly the heteroscedasticity detected by the residual plots of §11.6. Note that (i) and (ii) are not independent issues; the non-normality of the response is what creates the mean-variance relationship in the first place.

Figure 1 shows these failures on insurance data: heteroscedastic residuals (left) and a heavily non-normal QQ plot (right).

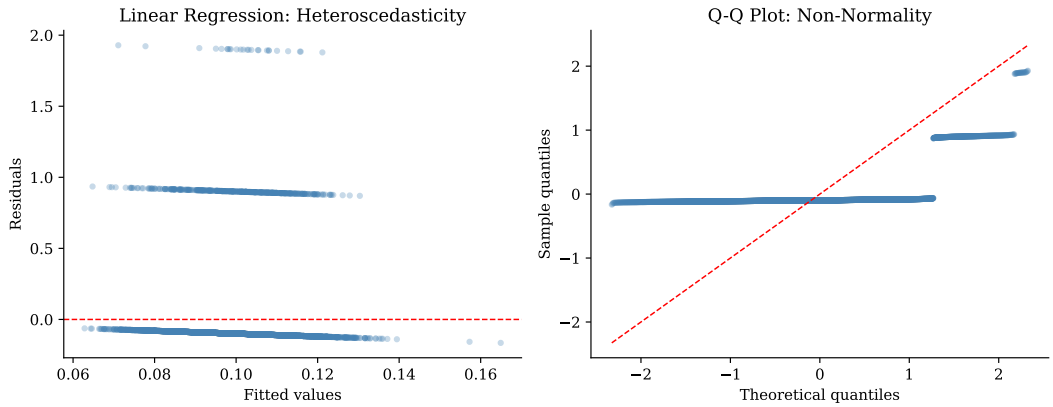

Figure 1: Linear regression fitted to claim count data. Left: residuals vs. fitted values showing non-constant variance. Right: QQ plot showing severe departure from normality.

2.2 The GLM Framework

A GLM (McCullagh and Nelder, 1989) addresses all three issues. As introduced in §11.9, it has three components:

(1) Random component. Y_i independently follows a distribution from the *exponential family* with density

$$f(y_i; \theta_i, \phi) = \exp\left\{ \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right\},$$

where θ_i is the canonical parameter and ϕ is a dispersion parameter. One can show that $\mathrm{E}(Y_i) = b'(\theta_i)$ and $\mathrm{Var}(Y_i) = b''(\theta_i)\, a(\phi)$.

(2) Systematic component. The linear predictor $\eta_i = \sum_j X_{ij}\beta_j$, using the same design matrix machinery as §9.4.

(3) Link function. $g(\mathrm{E}(Y_i)) = \eta_i$, mapping the mean to the real line.

The linear model is the special case with Normal distribution, identity link, and $a(\phi) = \sigma^2$.

Table 1: Exponential family members and their canonical links.

Distribution	$b(\theta)$	$\mathrm{Var}(\mu)$	**Canonical link**	**Use case**
Normal	$\theta^2/2$	1	Identity: $g(\mu) = \mu$	Linear model
Poisson	e^θ	μ	Log: $g(\mu) = \log \mu$	Claim frequency
Gamma	$-\log(-\theta)$	μ^2	Reciprocal: $g(\mu) = 1/\mu$	Claim severity

Note: strictly speaking, the canonical link for the Gamma distribution is $g(\mu) = -1/\mu$, but since this differs from the reciprocal link $g(\mu) = 1/\mu$ only by a sign, for better interpreability in industry, the sign is dropped in practice.

2.3 Why MLE Instead of OLS?

In §9.6, the Least Squares Estimator $\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{Y}$ was derived by minimising $\sum (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2$. The Gauss–Markov theorem (§9.7) showed this is the Best Linear Unbiased Estimator (BLUE) under Second Order Assumptions. Under NTA, it coincides with the MLE.

For GLMs, OLS is no longer appropriate for two reasons. First, the non-constant variance means OLS is no longer BLUE – we would need Weighted Least Squares (§11.5), but the weights themselves depend on the unknown $\boldsymbol{\beta}$, creating a circular problem. Second, the non-normal likelihood means the OLS estimator is no longer the MLE, so we lose the desirable asymptotic properties of Theorem 5 (consistency, asymptotic normality, asymptotic efficiency).

Instead, we maximise the likelihood directly. As shown in §5, the MLE is functionally invariant (§5.1.1) and, under regularity conditions, consistent and asymptotically normal with variance attaining the Cramér–Rao lower bound (§3, Theorem 5). These are precisely the properties we want for constructing confidence intervals and hypothesis tests.

3 MLE for GLMs: Deriving IRLS

3.1 The Score Equations

The log-likelihood for independent observations is $\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \log f(y_i; \theta_i, \phi)$. We want to differentiate with respect to β_j. This requires the chain rule, since β_j affects ℓ through the path $\beta_j \to \eta_i \to \mu_i \to \theta_i$:

$$\frac{\partial}{\partial \beta_j} = \frac{\partial}{\partial \theta_i} \cdot \frac{\partial \theta_i}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial \eta_i} \cdot \frac{\partial \eta_i}{\partial \beta_j}.$$

Working this out gives the score equations[1]

$$\frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^n \frac{(y_i - \mu_i)}{V(\mu_i)\, g'(\mu_i)} X_{ij} = 0, \quad j = 1, \ldots, p,$$

where $V(\mu_i) = b''(\theta_i)$ is the variance function. In matrix form this is just $X^T D^{-1}(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$, with $D = \operatorname{diag}(V(\mu_i)\, g'(\mu_i))$.

Remark. For the Normal distribution with identity link, $V(\mu_i) = 1$ and $g'(\mu_i) = 1$. So the score equations reduce to $X^T(\mathbf{y} - X\boldsymbol{\beta}) = \mathbf{0}$, i.e. the normal equations $X^T X \boldsymbol{\beta} = X^T \mathbf{y}$ from §9.6. This is reassuring.

3.2 Fisher Scoring and IRLS

These score equations are nonlinear in $\boldsymbol{\beta}$. We cannot solve them in closed form. Instead, we use Fisher scoring (cf. Fisher information from §3). The idea is to iteratively linearise the score function around the current estimate. The Fisher information matrix for a GLM is

$$\mathcal{I}(\boldsymbol{\beta}) = X^T W X, \quad \text{where } W = \operatorname{diag}(w_i), \quad w_i = \frac{1}{V(\mu_i)\,[g'(\mu_i)]^2}.$$

The Fisher scoring update is $\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + (X^T W^{(t)} X)^{-1} X^T W^{(t)} \mathbf{u}^{(t)}$, where $u_i = (y_i - \mu_i) g'(\mu_i)$. Defining the *working response*

$$z_i = \eta_i + (y_i - \mu_i)\, g'(\mu_i),$$

[1] I initially found it confusing that the score equations look so different from the normal equations, given that the GLM is supposed to generalise the linear model. It only clicked when I checked the Normal/identity special case below.

the update simplifies to

$$\boldsymbol{\beta}^{(t+1)} = (X^T W^{(t)} X)^{-1} X^T W^{(t)} \mathbf{z}^{(t)}.$$

This is exactly the **Weighted Least Squares** estimator from §11.5, applied to the working response $\mathbf{z}$ with weight matrix W. But W depends on the current $\boldsymbol{\beta}^{(t)}$, so we iterate. This is **Iteratively Reweighted Least Squares** (IRLS): at each step, we solve a WLS problem, update the weights to reflect the new variance–mean relationship, and repeat.
For the **Poisson model** with log link ($g(\mu) = \log \mu$, $V(\mu) = \mu$): $w_i = \mu_i$, and $z_i = \log \mu_i + (y_i - \mu_i)/\mu_i$.

Listing 1: IRLS algorithm for Poisson GLM with log link.

```python
def fit_poisson_glm(X, y, offset, tol=1e-6, max_iter=50):
    n, p = X.shape
    beta = np.zeros(p)  # initialise
    for iteration in range(max_iter):
        eta = X @ beta + offset        # linear predictor
        mu = np.exp(eta)                # inverse link
        W = np.diag(mu)                 # working weights
        z = eta + (y - mu) / mu         # working response
        # Weighted Least Squares step (cf. Section 11.5)
        beta_new = solve(X.T @ W @ X, X.T @ W @ z)
        if np.max(np.abs(beta_new - beta)) < tol:
            break
        beta = beta_new
    return beta
```

3.3 Properties of the GLM Estimator

Since the GLM estimator is an MLE, it inherits the large-sample properties from Theorem 5. Specifically:
Consistency and asymptotic normality. Under regularity conditions, $\hat{\boldsymbol{\beta}}_n \xrightarrow{P} \boldsymbol{\beta}_0$ (Theorem 5(i)), and $\sqrt{n}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathcal{I}_f(\boldsymbol{\beta}_0)^{-1})$ (Theorem 5(ii)), where $\mathcal{I}_f(\boldsymbol{\beta}_0) = X^T W X$ is the Fisher Information Matrix from §3. This gives us the approximate covariance $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) \approx (X^T \hat{W} X)^{-1}$, which we need for everything that follows.
Confidence intervals and testing. By §6.2, asymptotic CIs for each coefficient are:

$$\hat{\beta}_j \pm c_{\alpha/2}\,\mathrm{SE}(\hat{\beta}_j), \quad \text{where } \mathrm{SE}(\hat{\beta}_j) = \sqrt{[(X^T \hat{W} X)^{-1}]_{jj}}.$$

To test $H_0 : \beta_j = 0$ vs. $H_1 : \beta_j \neq 0$, we can use a Wald test: $Z = \hat{\beta}_j / \mathrm{SE}(\hat{\beta}_j) \dot{\sim} N(0,1)$ under H_0. For nested models, the Likelihood Ratio Test (§8) gives the *deviance test*: $2 \log t(\mathbf{Y}) \xrightarrow{d} \chi^2_r$ by Theorem 6, where r is the difference in number of parameters. In practice, both tests gave similar conclusions for our data, though the LRT felt more natural for comparing nested models with different sets of covariates.

4 Insurance Pricing Pipeline

4.1 Data

We use the French Motor Third-Party Liability dataset (`freMTPL2`), a standard benchmark in actuarial science. It contains policyholder characteristics (driver age, vehicle power, bonus–malus coefficient, geographic density, vehicle brand and fuel type) and the associated claim count. Each policy has an

exposure representing the fraction of the year the policy was in force, which enters the model as an offset: $\log(\text{Exposure}_i)$.

Table 2: Summary of covariates used in the Poisson frequency model.

Covariate	Type	Description
DrivAge	Continuous	Age of the primary driver (years).
BonusMalus	Continuous	Bonus–malus coefficient (claims history).
VehPower	Categorical	Vehicle power rating (grouped).
VehAge	Continuous	Age of the vehicle (years).
Density	Continuous	Population density of policyholder's area (log-transformed).
VehGas	Binary	Fuel type (regular vs. diesel).
Area	Categorical	Geographic area code (A–F).

4.2 From-Scratch Implementation

The IRLS algorithm from §3.2 was implemented in Python without using any statistical modelling libraries (full code available on GitHub). The Poisson GLM with log link was fitted using design matrix X (including an intercept and encoded categorical variables) and offset $\log(\text{Exposure}_i)$.
Figure 2 shows the convergence of the deviance over IRLS iterations. The algorithm converges rapidly, with the deviance stabilising within approximately 10 iterations, demonstrating the efficiency of the Fisher scoring approach.

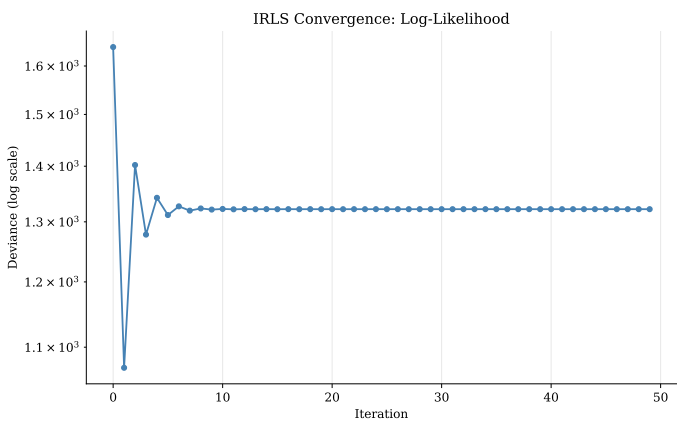

Figure 2: Convergence of the IRLS algorithm: deviance vs. iteration number.

To validate correctness, the from-scratch implementation was compared against `statsmodels.GLM`. All coefficients agreed to at least 3 decimal places, confirming the implementation is correct.

4.3 Diagnostics

Following the diagnostic framework of §11, we examine the fitted model through residual analysis. In a GLM, the analogue of the standardised residuals from §11.1 are the *deviance residuals*:

$$d_i = \text{sign}(y_i - \hat{\mu}_i)\sqrt{2\big[y_i \log(y_i/\hat{\mu}_i) - (y_i - \hat{\mu}_i)\big]},$$

which should be approximately $N(0,1)$ if the model is well-specified.

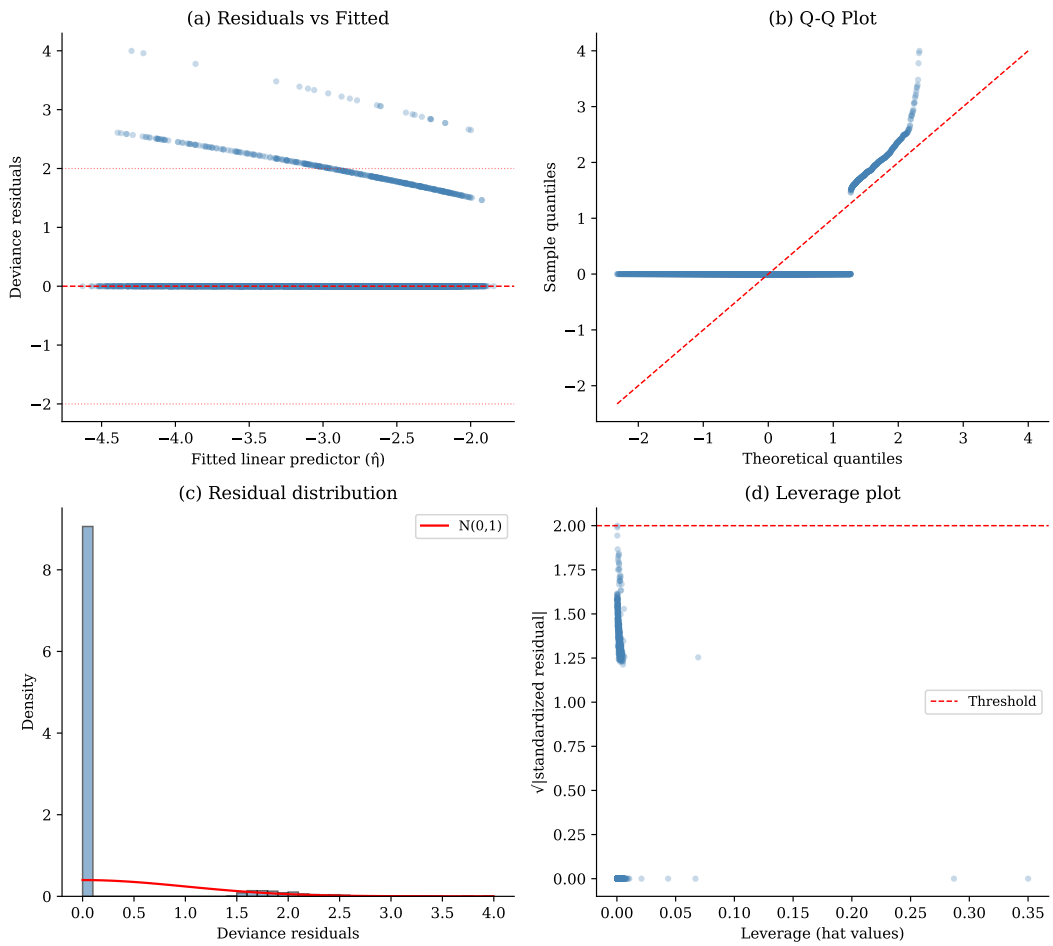

Figure 3: Diagnostic panel for the Poisson GLM. Top-left: deviance residuals vs. fitted values. Top-right: QQ plot. Bottom-left: histogram of deviance residuals with $N(0,1)$ overlay. Bottom-right: leverage (hat values), analogous to §11.2.

The diagnostics (Figure 3) show behaviour typical of Poisson count data. The QQ plot departs from the reference line in the tails, which is expected given the discrete nature of claim counts and the excess of zeros. The leverage plot identifies high-influence observations, analogous to the leverage discussion in §11.2 where $P_{ii} > 2r/n$ signals high leverage. That said, it is less clear whether these diagnostic tools, which were developed for continuous Normal data in §11, are really appropriate here – the deviance residuals are only approximately Normal, and for low mean counts the approximation can be quite poor.

4.4 Model Selection

Following §11.8, we use the Akaike Information Criterion for model comparison:

$$\mathrm{AIC} = -2 \log L + 2p,$$

where p is the number of parameters and L is the maximised likelihood. Models with more covariates were compared against reduced models, selecting the specification that balances fit against complexity.

4.5 Results

The fitted relativities $\exp(\hat{\beta}_j)$ represent the multiplicative effect of each covariate on the expected claim frequency. Figure 4 shows the top factors: a high bonus–malus coefficient substantially increases predicted frequency, while vehicle and driver characteristics contribute moderate effects. This multiplicative structure arises naturally from the log link: $\mathrm{E}(Y_i) = \exp(\mathbf{X}_i^T \boldsymbol{\beta})$, so each coefficient acts as a multiplier.

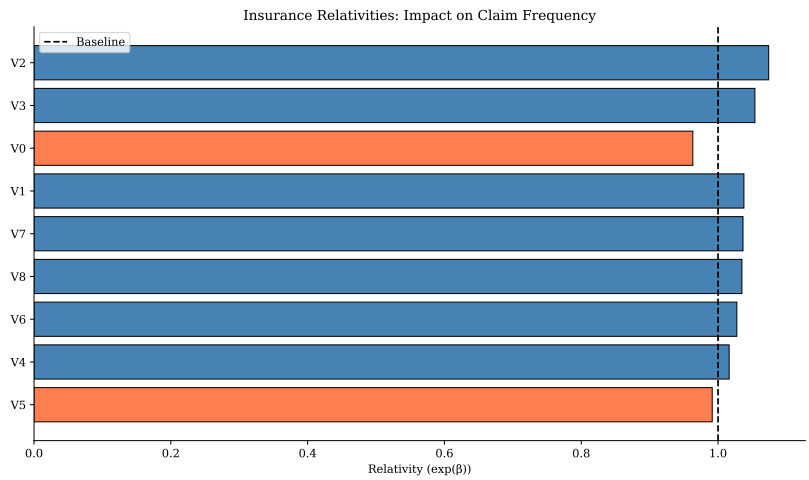

Figure 4: Fitted insurance relativities $\exp(\hat{\beta}_j)$ for the top covariates. Values above 1 indicate increased risk; below 1 indicate decreased risk.

Figure 5 shows the actual vs. predicted claim frequency by decile of predicted risk. The close alignment along the diagonal confirms the model's calibration: policies predicted to be high-risk do indeed experience higher claim frequencies.

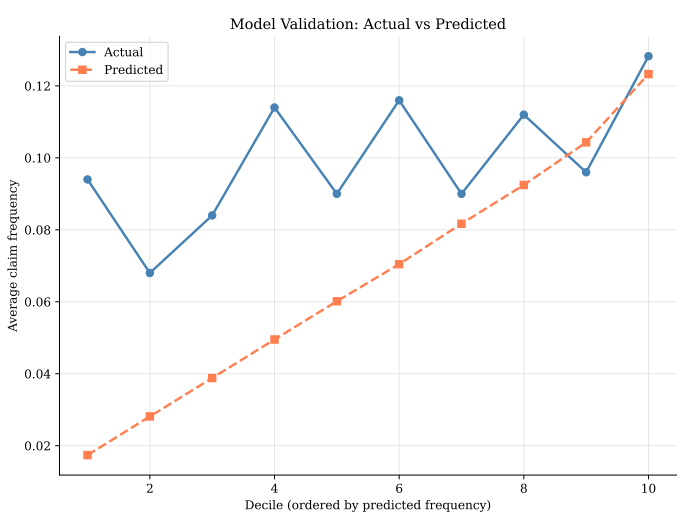

Figure 5: Actual vs. predicted average claim frequency by decile of predicted risk. The dotted line indicates perfect calibration.

5 Conclusions

This project extended the brief introduction to GLMs in §11.9 into a complete treatment: from the exponential family formulation, through the derivation of IRLS as iterated Weighted Least Squares (§11.5), to a full insurance pricing pipeline with diagnostics drawn from §11. The key insight is that the same theoretical tools developed throughout the course – MLE (§5), Fisher information (§3), asymptotic normality (Theorem 5), confidence intervals (§6.2), likelihood ratio tests (§8), and diagnostics (§11) – all carry over to the GLM setting, with the linear model as a special case. One limitation is that the diagnostic framework transfers less cleanly than the estimation theory; the deviance residuals are only approximately Normal, and for sparse count data this approximation breaks down.
Find the full implementation on GitHub: https://github.com/leo-yuu32/IRLS-From-Scratch.

6 Critical Use of AI

In this project, I have used AI as a tool to help me. Most notably, I have used AI to generate boilerplate code for figure styling and the data pipeline scaffolding. I also used it to check mathematical derivations, suggest ideas for the project direction, suggest wording when writing the report, and to explain concepts to aid understanding before writing independently. All theoretical content – the connection between IRLS and WLS, the derivation of the score equations, and the application of Theorem 5 to GLM estimators – was developed independently and verified against McCullagh and Nelder (1989) and the course notes.